CSC10014 · MID-TERM PRESENTATION

NutriSnap

Snap your meal. Know your nutrition. Build healthier habits.

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Meet our team

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6 members · CSC10014 Computational Thinking · Faculty of Information Technology, HCMUS

TOPIC 01 · PRODUCT OVERVIEW

The story behind NutriSnap

Who we are building for – and why



Meet Minh Anh – a story many students know

Day 1



"I will eat healthier this month!"

Downloads a calorie app, full of motivation.

Day 3



"Where is cơm tấm in this list?"

Searches every dish, guesses grams, types numbers.

Day 5



"This takes too long..."

Stops logging. The goal is forgotten.

Our mission

Make healthy tracking take seconds, not minutes – with Vietnamese food first and numbers you can trust.

Persona based on our own experience as students.

Who NutriSnap is designed for



The student

GOAL

Eat healthy on ~50,000 VND a day

NUTRISNAP GIVES

Budget-aware dish suggestions, simple daily target



The office worker

GOAL

Lose weight safely, without strict diets

NUTRISNAP GIVES

Safe weekly pace, weight trend, reminders



The gym-goer

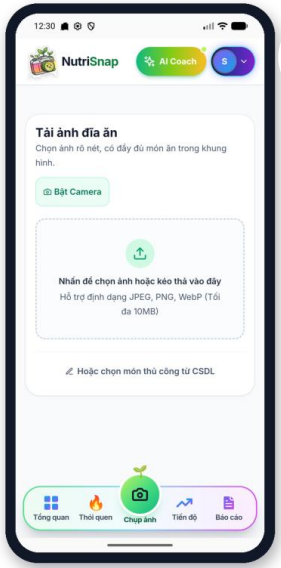
GOAL

Enough protein to build muscle

NUTRISNAP GIVES

Macro targets, protein-rich food swaps

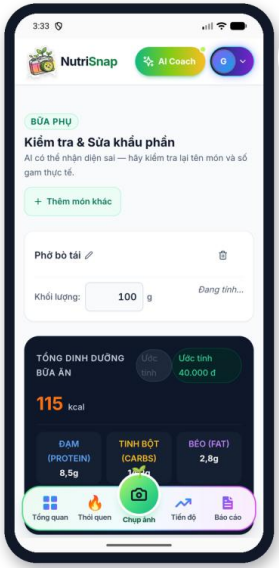
Meet NutriSnap: Snap → Confirm → Coach



1

Snap

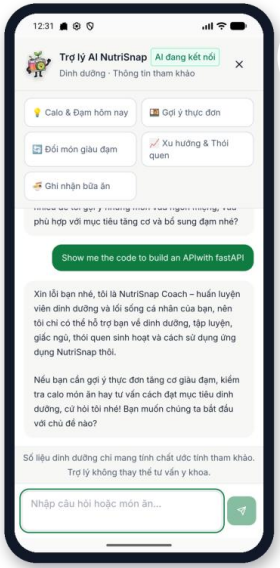
Take a photo – the app recognises the dishes



2

Confirm

Check dishes and grams – nothing is saved before you agree



3

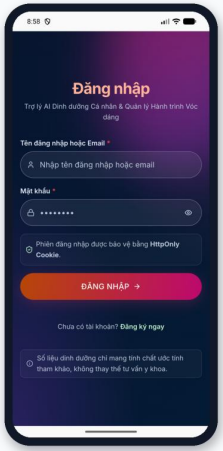
Coach

Ask anything – answers use your own data

One loop, every day: plan your goal → log your meal → see your progress → get coached.

User flow: from opening the app to getting advice

1 Open & log in



Personal goal ready

2 Snap / log food

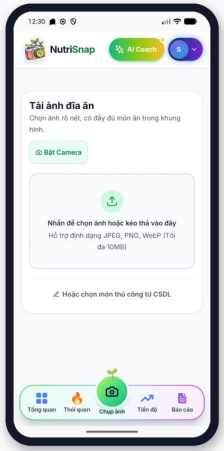
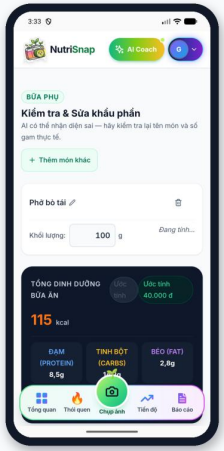


Photo or search

3 Confirm meal



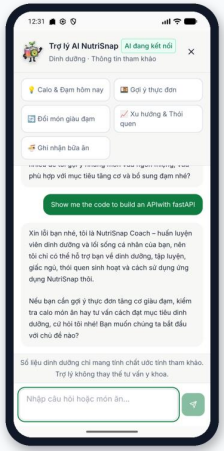
Check dishes & grams

4 See progress



Calories, macros, streak

5 Get advice



Coach & food swaps

Core features and their status



Working

Food logging

Photo or manual search



Working

Nutrition & calories

Daily ring + 3 macros



Working

Goals & profile

BMR, TDEE, safe pace



Working

Suggestions

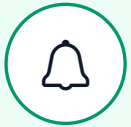
AI coach & food swaps



Working

Habits & reports

Streaks, badges, monthly PDF



Working

Meal reminders

Lock screen, with mascots

TOPIC 02 · MARKET

Where NutriSnap stands

Real user problems and how we are different



Target market and user problems

Target market

- Young Vietnamese smartphone users
- University students on a budget
- Health-conscious people tracking daily food
- People who eat mostly Vietnamese dishes

Numbers are confusing

Labels and calories are hard to understand

Manual logging is tiring

Search, guess, type – for every meal

Needs are unknown

Few people know their daily calorie or protein needs

Local food is missing

Mixed Vietnamese dishes are rare in global apps

How we compare

	MyFitnessPal	YAZIO	Cronometer	NutriSnap
Main strength	Huge food database	Clean diet plans	Micronutrient detail	Photo + local data + coach
How you log	Search / barcode	Search / barcode	Search, detailed	Photo first, then confirm
Vietnamese dishes	User-added, varies	Limited	Limited	~500 curated dishes
AI coach on your data	Limited	Limited	No	Yes, in Vietnamese & English
Price	Freemium	Freemium	Freemium	Free (student project)

Team assessment from public product descriptions – competitor features and prices may change.

Our positioning: what sets us apart

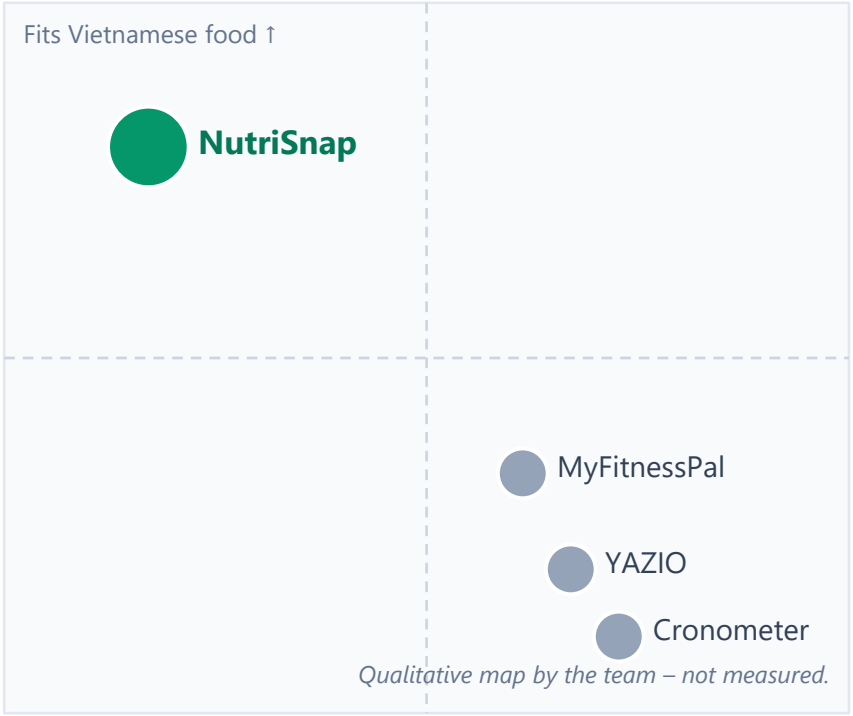


Photo-first, confirm-always

Logging in seconds, but the user stays in control

Vietnamese + global data

~500 curated Vietnamese dishes plus 13,159 USDA foods

A coach that knows you

Answers from your own goals and meals, stays on topic

TOPIC 03 · ENGINEERING

UI/UX and system plan

Real screens, one design system, a clear path to code



Design system and consistency

Colours

- Primary green
- Fresh lime
- Warm accent
- Info blue
- Highlight
- Ink / dark mode



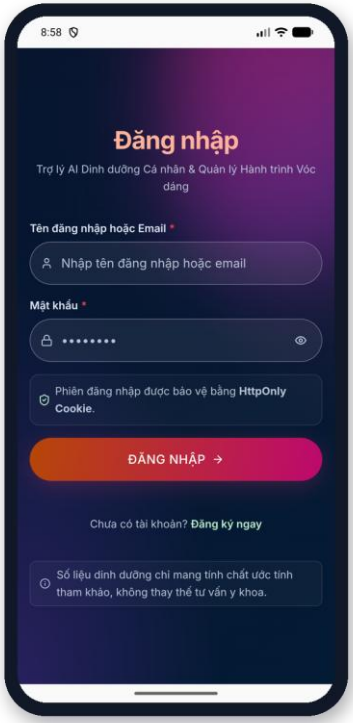
Consistency rules

- Same header and bottom bar on every screen
- Raised camera button always in the centre
- Rounded cards, big numbers, short labels
- Light and dark mode share one layout
- One friendly mascot in eight moods

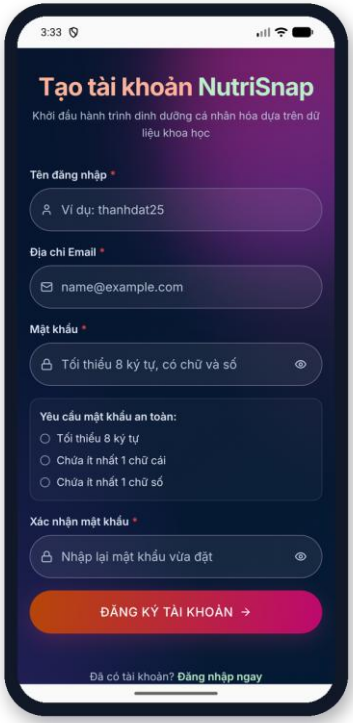


Same screen, light and dark

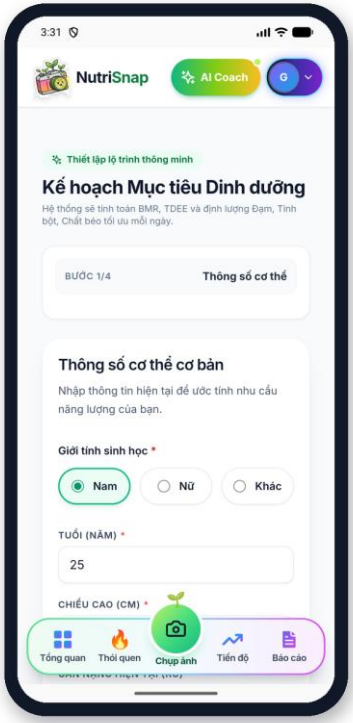
UI · Onboarding: sign up and set a personal goal



Login



Register

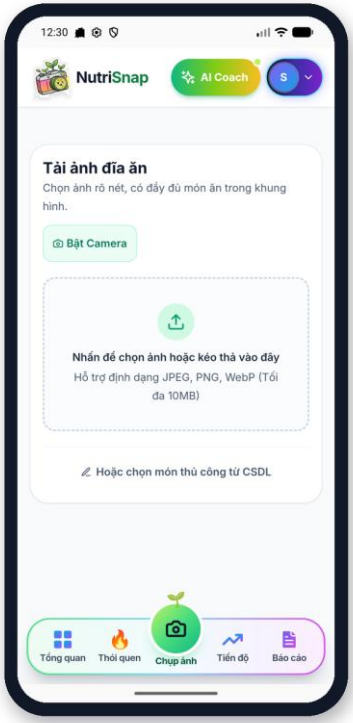


Goal wizard · step 1/4

UX decisions

- Login and register share one visual style
- Live password checklist while typing
- Goal wizard: 4 short steps, one topic each
- Safe weekly pace shown before saving

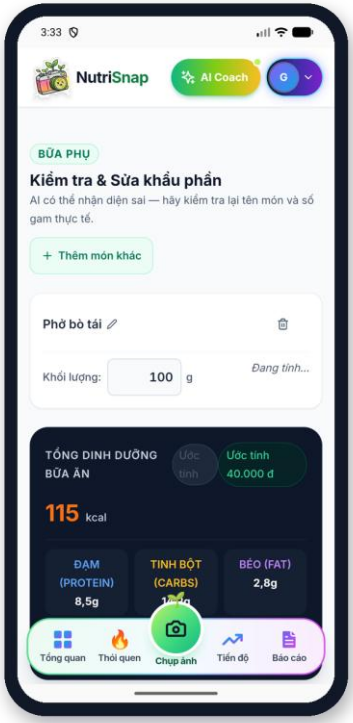
UI · Core loop: snap, search and confirm a meal



Camera / upload



Manual search

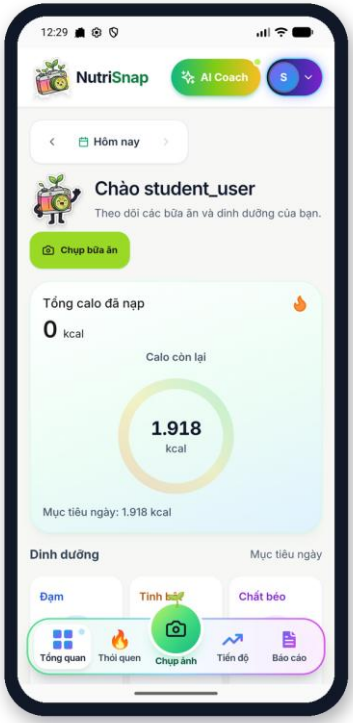


Review & edit grams

UX decisions

- Two ways in: photo or manual search
- Meal type chosen with one tap
- Grams are editable; totals update
- Non-food photo → “please retake”
- Nothing saved until you confirm

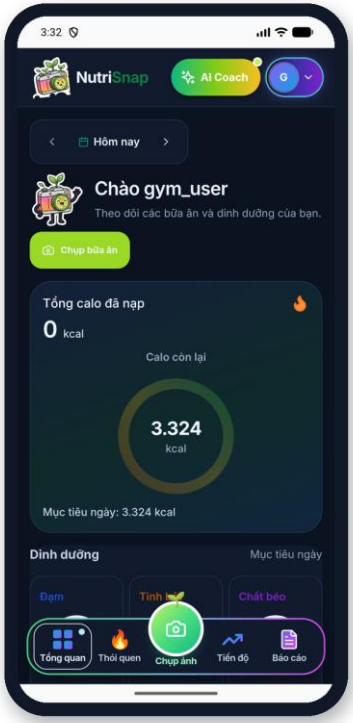
UI · Daily dashboard: today at a glance



Calories ring



Macros & meals

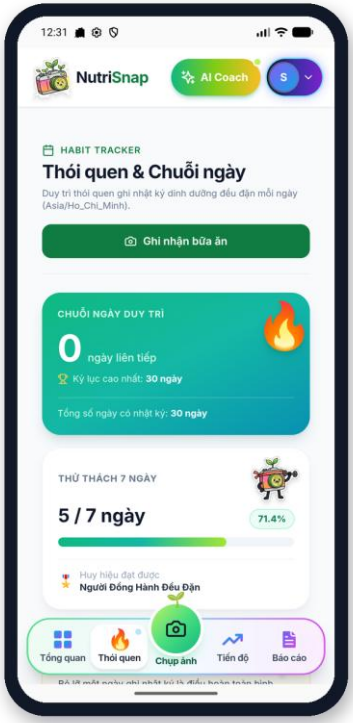


Dark mode

UX decisions

- One big number: calories left
- Protein, carbs, fat as simple rings
- Mascot greets and encourages
- Friendly empty state with a clear next step

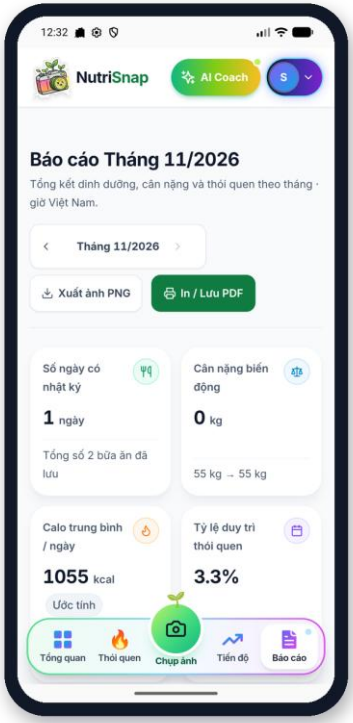
UI · Habits, progress and reports



Streaks & challenge



Weight trend

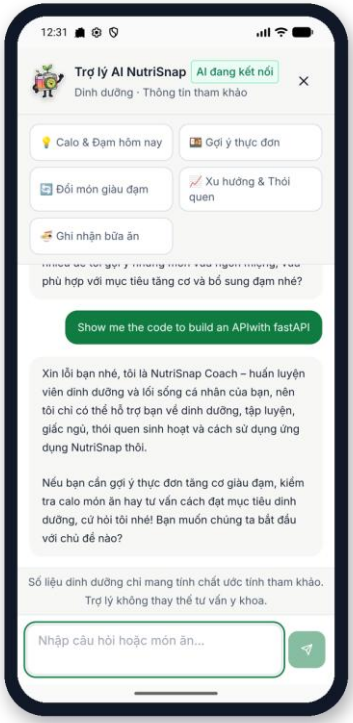


Monthly report

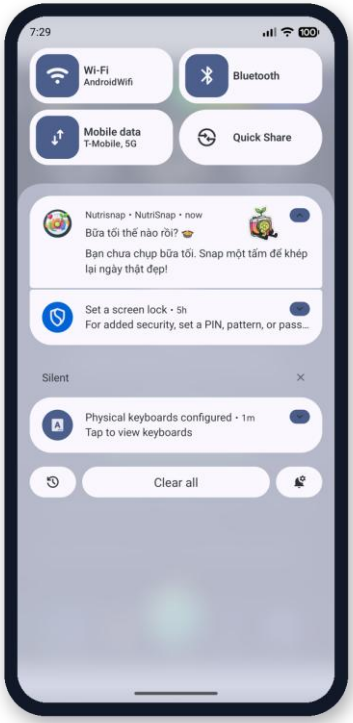
UX decisions

- Streak, 7-day challenge and badges
- Trend over 7 / 14 / 30 / 90 days
- Calm colours – no alarming red
- Export report as PNG or PDF

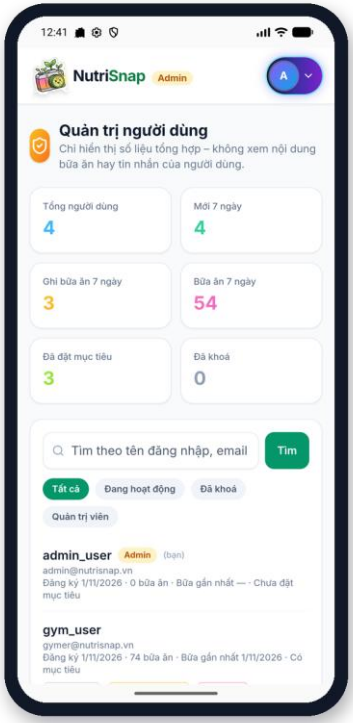
UI · AI coach, reminders and admin



AI coach



Meal reminder



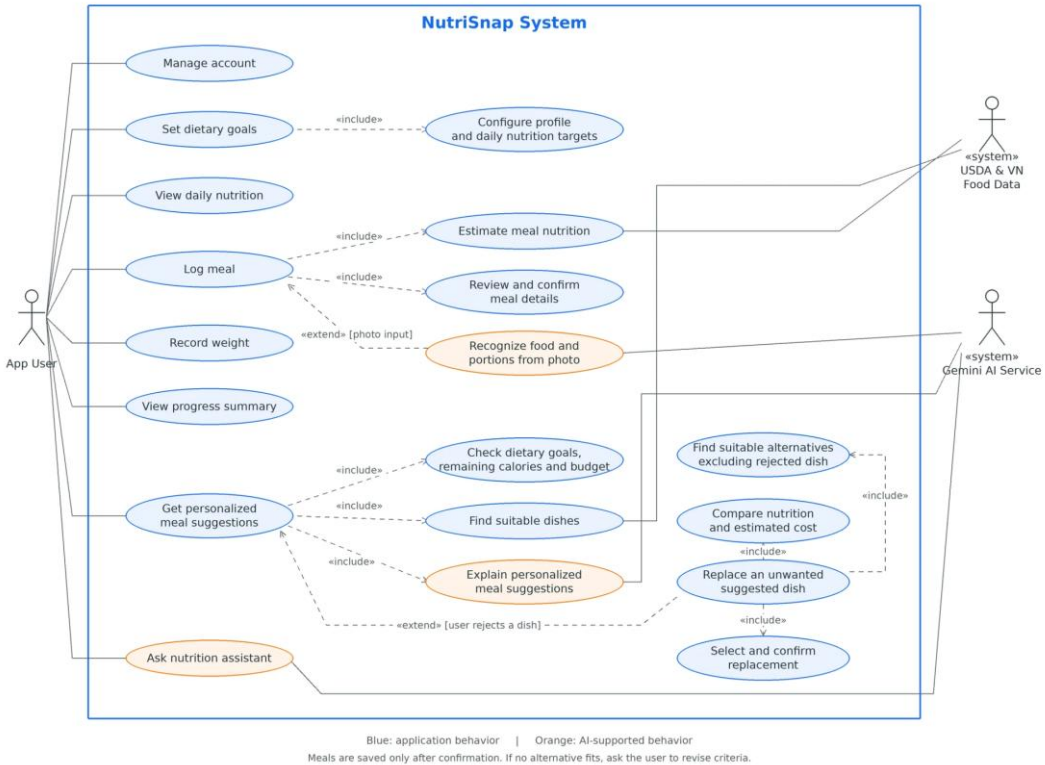
Admin panel

UX decisions

- Coach is one tap away in the header
- Quick-question chips to start
- Politely refuses off-topic requests
- Reminders with mascots on the lock screen
- Admin has its own separate panel

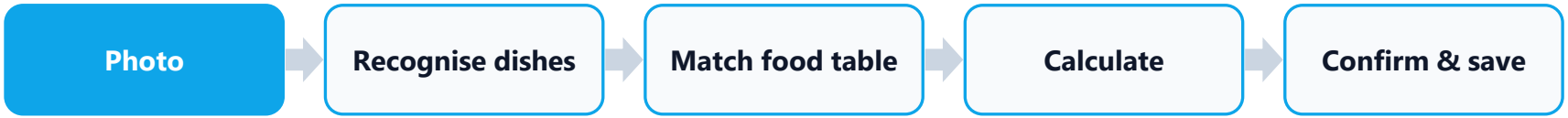
Workflow: function list and use case diagram

- **F1** Manage account
- **F2** Set dietary goals
- **F3** Log meal (photo / manual)
- **F4** Review & confirm meal
- **F5** View daily nutrition
- **F6** Record weight
- **F7** Habits & progress
- **F8** Monthly report
- **F9** Ask AI coach / swap food
- **F10** Meal reminders



Core pipelines: how the 3 key functions work

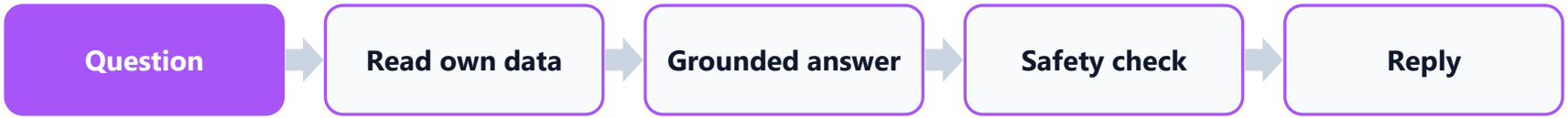
1 Snap to nutrition



2 Personal goal planner

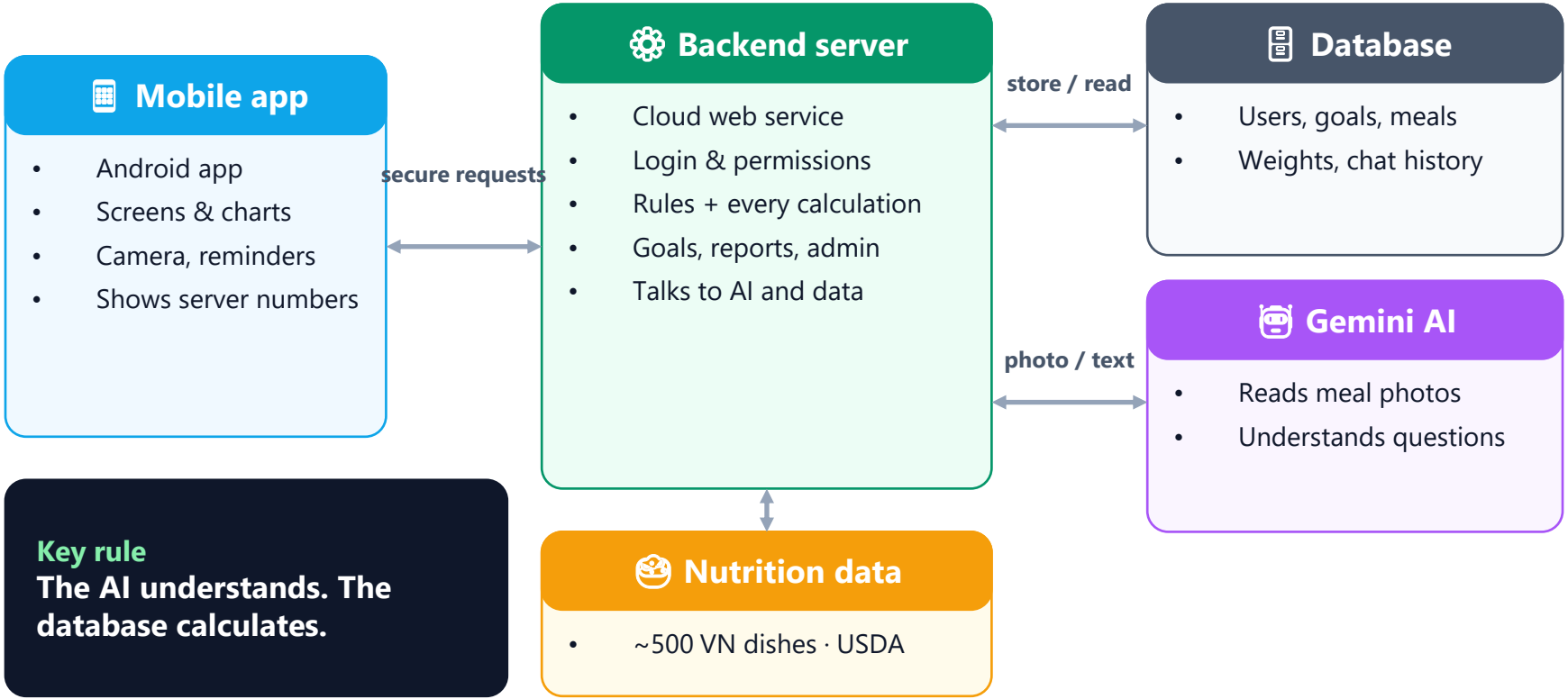


3 AI coach



Rule in every pipeline: the AI helps to understand – all numbers come from the food database.

System architecture



System overview: how it all works

Frontend

Mobile app (Android)

- Shows screens and charts
- Camera and reminders
- Never calculates nutrition itself

Backend

Server in the cloud

- Business rules and every calculation
- Goal planner and nutrition engine
- Reports, habits, admin

Data

Where numbers come from

- ~500 Vietnamese dishes
- USDA FoodData Central
- Cloud database for user data

AI

Understands, does not count

- Reads meal photos
- Understands questions
- Answers grounded in user data

Security

Your data stays yours

- Protected login cookies
- Each user sees only own data
- Admin separated from users

Flow: app → secure request → server rules → food data + AI → result back to the app

How NutriSnap will grow – and thank you

Design	Build	Test	Improve	Release
UI, functions, use case	Working prototype	Real users & accuracy	Barcode, water tracker	Public Android app
Done	**Done**	**Next**	**Next**	**Planned**

Our coding approach

- Small feature branches, reviewed before merging
- Build each function end to end: screen → server → data
- Keep one design system for every new screen
- Test with real users after each step

Thank you!



Questions and feedback are welcome.